Software Engineering

Term Project Report– Fall 2023

DIGITAL CANTEEN BILLING SYSTEM

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Acknowledgement

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"Not every teacher deserves to be called one because that title reserved for great ones like you!"

"An amazing teacher like you is hard to find and impossible to forget!"

"Give your teachers the respect they deserve, because they are the ones who can help you get where you need to go!"

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Description Of The System

Overview:

This software aims to streamline the operations of canteens or small shops by automating the billing process, managing inventory, and tracking borrowed items. It is designed to make transactions faster, more accurate, and easily accessible for both the shopkeeper and customers.

Key Features:

- ✓ **User-Friendly Interface:** Develop an intuitive interface designed for ease of use by the shopkeeper.
- ✓ **Product Database:** Store a database of available items with their prices.
- ✓ **Quick Billing:** Enable quick selection of items and generate bills instantly.
- ✓ **Transaction Records**: Maintain a digital record of transactions, including date, time, items purchased, and total amount.
- ✓ **Borrowing System**: Allow customers to borrow items, recording their details and the borrowed items.
- ✓ **Return Reminders**: Send notifications or reminders to customers about returning borrowed items.
- ✓ **Payment Options**: Support multiple payment methods like cash, card, mobile payments, etc.
- ✓ **Receipt Generation**: Provide printable or digital receipts for customers.
- ✓ **Reporting and Analytics**: Generate reports on daily, weekly, or monthly sales to help the shop owner track, trends and manage inventory.
- ✓ **Security**: Implement secure login systems and data encryption to safeguard sensitive information.
- ✓ **Compatibility**: Ensure compatibility across devices like tablets, desktops, or mobile phones.

Benefits:

- ✓ **Efficiency**: Speed up the billing process, reducing waiting times for customers.
- ✓ **Accuracy**: Minimize errors in calculations and recording transactions.
- ✓ **Record-Keeping**: Maintain a comprehensive digital record for accounting and auditing purposes.
- ✓ **Customer Satisfaction**: Provide a smoother and more professional checkout experience.

Scope:

- ➢ The proposed comprehensive canteen management system aims to modernize and simplify routine operations by digitizing billing, inventory management, and tracking borrowed items. This holistic solution benefits both the shop management and customers by offering efficiency, accuracy, and improved service.

Functional Requirements:

1. **User Authentication:**

    - *Description:* Users (both shopkeepers and customers) should have secure login credentials.
    - *Purpose:* To ensure data security and restrict unauthorized access.

2. **Billing System:**

    - *Description:* Allow shopkeepers to select items, calculate totals, and generate bills.
    - *Purpose:* Streamline the checkout process for faster transactions.

3. **Transaction Records:**

    - *Description:* Record details of each transaction, including date, time, items purchased, and total amount.
    - *Purpose:* Maintain a digital ledger for accounting and auditing purposes.

4. **Inventory Management:**

    - *Description:* Keep track of available stock and send alerts for items that need restocking.
    - *Purpose:* Ensure sufficient stock levels and prevent stockouts.

5. **Borrowing System:**

    - *Description:* Allow customers to borrow items, recording their details and the borrowed items.
    - *Purpose:* Enable customers to borrow items with a clear record for tracking.

6. **Return Reminders:**

    - *Description:* Send notifications or reminders to customers about returning borrowed items.
    - *Purpose:* Ensure timely return of borrowed items and efficient inventory management.

7. **Payment Options:**

    - *Description:* Support multiple payment methods such as cash, card, and mobile payments.
    - *Purpose:* Provide flexibility for customers and streamline the payment process.

8. **Receipt Generation:**

    - *Description:* Generate printable or digital receipts for customers after transactions.
    - *Purpose:* Provide customers with a record of their purchases for reference.

9. **Reporting and Analytics:**

    - *Description:* Generate reports on daily, weekly, or monthly sales, inventory levels, and borrowed items.
    - *Purpose:* Provide insights for business decision-making and trend analysis.

10. **Security Measures:**

    - *Description:* Implement secure login systems and data encryption.
    - *Purpose:* Ensure the security and integrity of user data.

Non-Functional Requirements:

1. **Usability:**

    - *Description:* The system should have an intuitive and user-friendly interface.
    - *Purpose:* Enhance user adoption and minimize training needs.

2. **Performance:**

    - *Description:* The system should handle multiple transactions simultaneously without significant delays.
    - *Purpose:* Ensure smooth and efficient operation during peak times.

3. **Scalability:**

    - *Description:* The system should be scalable to accommodate future growth in terms of users and data.
    - *Purpose:* Ensure the system can handle increased usage and data volume.

4. **Reliability:**

    - *Description:* The system should be reliable, with minimal downtime.
    - *Purpose:* Ensure continuous availability for shop operations.

5. **Compatibility:**

    - *Description:* The system should be compatible with various devices, such as tablets, desktops, and mobile phones.
    - *Purpose:* Ensure flexibility and accessibility for users.

6. **Data Backup and Recovery:**

    - *Description:* Implement regular data backup procedures and a reliable recovery mechanism.
    - *Purpose:* Prevent data loss and ensure quick recovery in case of system failures.

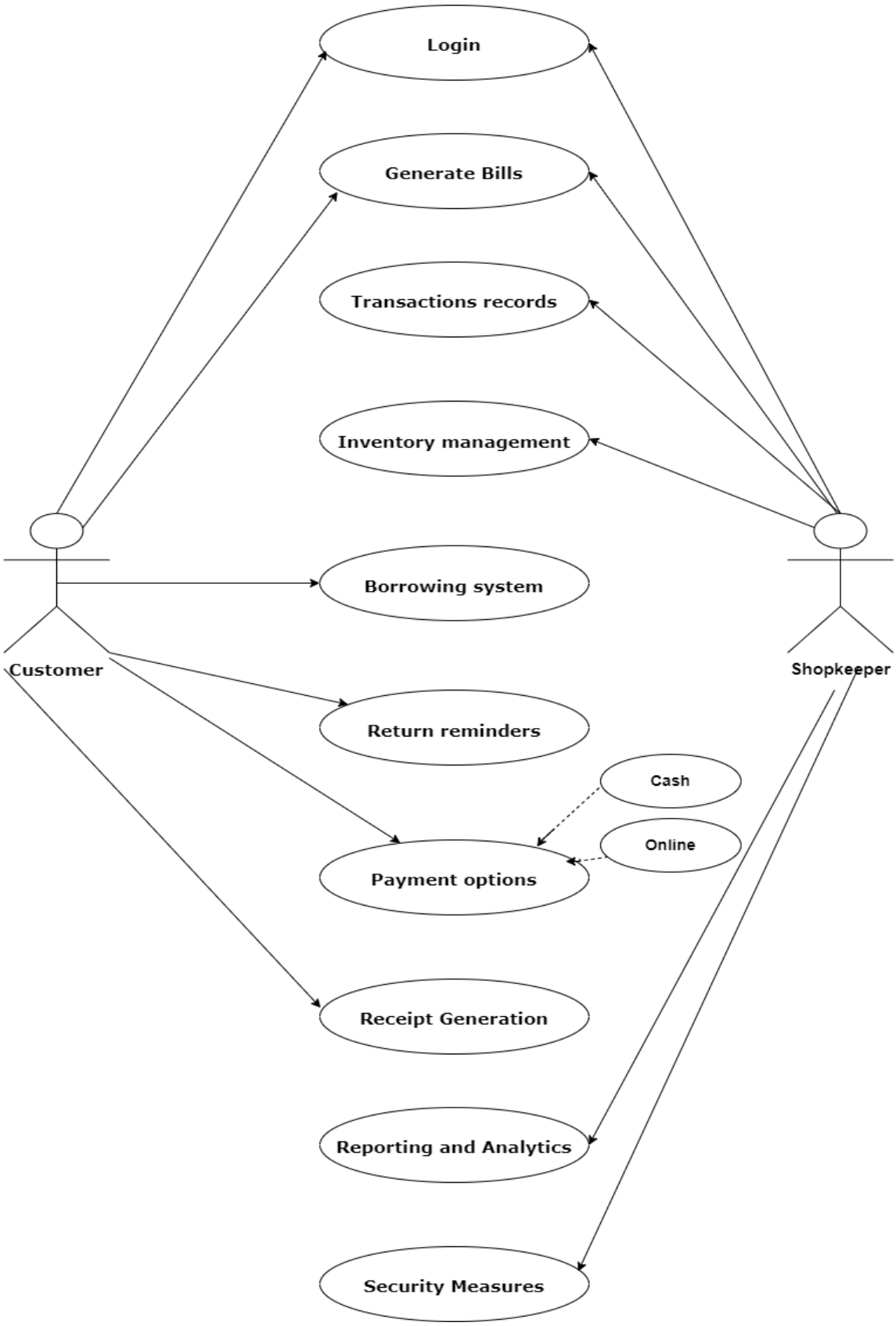
7. **Security Compliance:**

    - *Description:* Adhere to industry standards for data security and privacy.
    - *Purpose:* Ensure the system complies with legal and regulatory requirements.

8. **Training and Support:**

    - *Description:* Provide comprehensive training for shopkeepers and ongoing support.
    - *Purpose:* Facilitate smooth adoption and address any issues promptly.

USE CASE DIAGRAM:



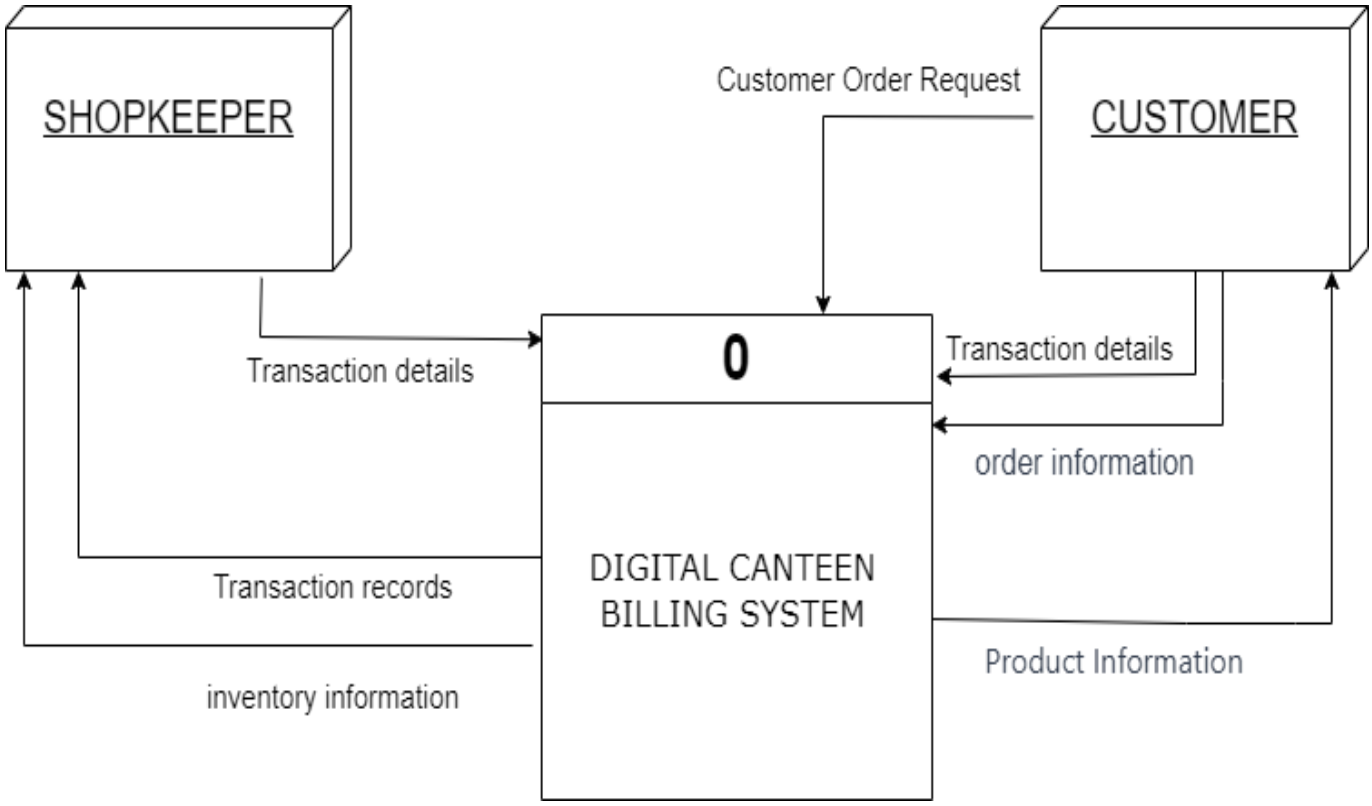
META DATA:

Use Case Number 01	Login	
ID:	1.1	
Actors Involved:	Customer, Shopkeeper	
Brief Description:	Ensures secure access for users (both shopkeepers and customers) through login credentials. Guarantees data security and prevents unauthorized access to the system.	
Pre-Conditions:	**System Availability:** The system must be operational and available for users to access the login interface. **User Registration:** Users (both shopkeepers and customers) must complete a registration process to obtain valid login credentials. **Secure Storage of Credentials:** The system must securely store user login credentials using encryption or other secure methods.	
Post-Conditions:	**Successful Authentication:** If the entered credentials are valid, the user gains access to the system. **Failed Authentication Attempts:** After a certain number of unsuccessful login attempts, the system may temporarily lock the user account or implement additional security measures**.** **Logout Option:** Users have the option to log out, terminating their session.	
Normal Flow of Events:	**Actor Action**	**System Response**
	1.The user navigates to the login interface. 2.The user enters the registration/ login username and password.	1.The system displays a login interface. 2.The system accepts the entered credentials as input for verification.

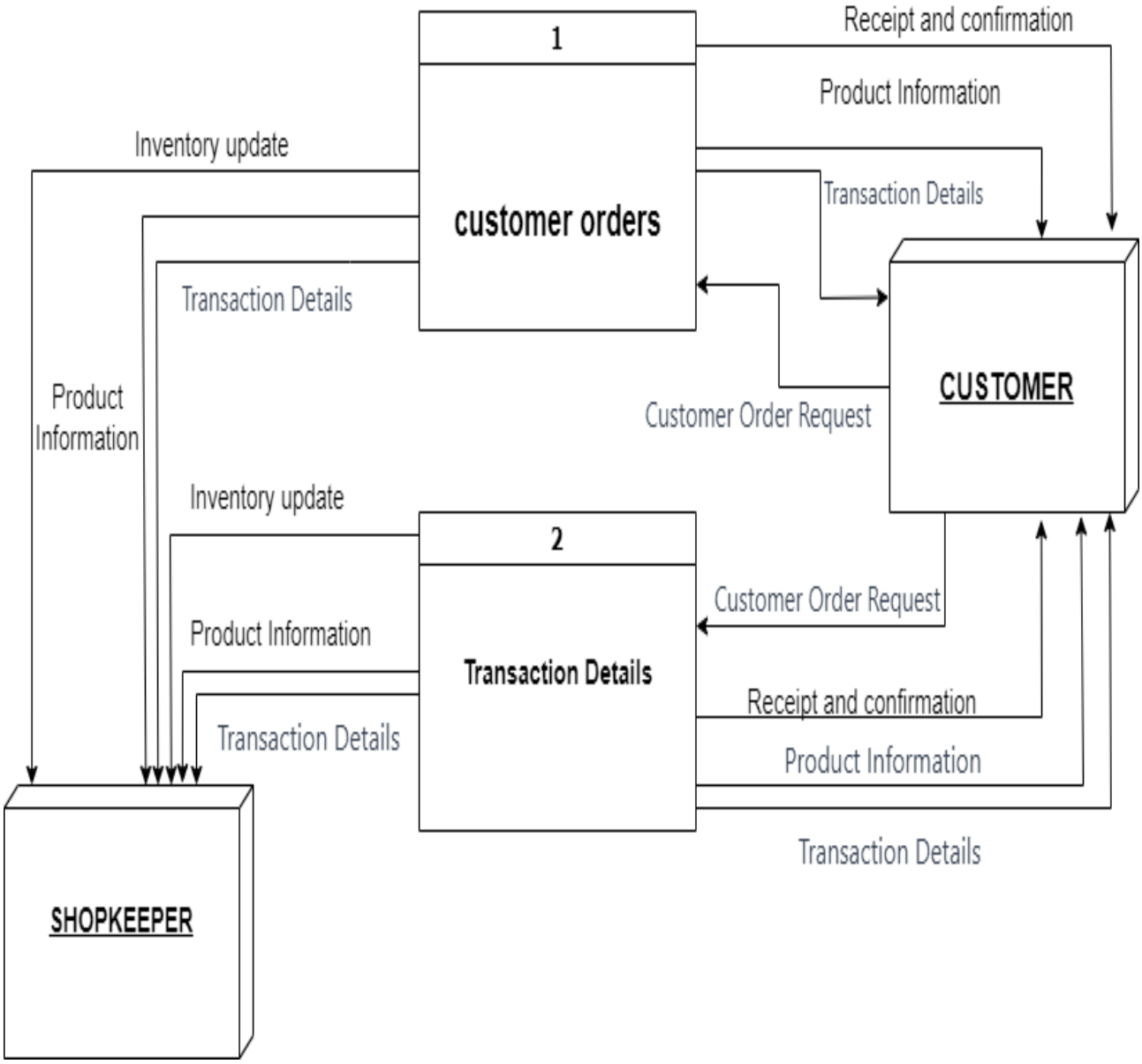
Use Case Number 02	Billing System	
ID:	1.2	
Actors Involved:	Customer, Shopkeeper	
Brief Description:	For shopkeepers, it facilitates the selection of items, calculation of totals, and generation of bills, streamlining the checkout process for faster and more efficient transactions. On the customer side, the system allows them to view selected items, confirm quantities, and receive detailed bills.	
Pre-Conditions:	**Product Database**: The system must have an up-to-date product database with accurate pricing information. **User Authentication**: Shopkeepers or users must be authenticated and logged into the system to access the application. **Items Selected:** 1.Shopkeepers need to have items selected by customers before initiating the billing process. 2.Customers select items for online orders.	
Post-Conditions:	**Transaction Record Creation:** After successful billing, a transaction record is created, including details such as items purchased, quantities, total amount, date, and time. **Inventory Update:** The system updates the inventory to reflect the sale, decrementing the quantities of sold items. **Receipt Generation**: A receipt is generated for the customer, summarizing the transaction details.	
Normal Flow of Events:	**Actor Action**	**System Response**
	1.Shopkeepers or customers choose products. 2.Shopkeepers or customer wants to generate the bills. 3.After completing a transaction, the user selects the option to generate a receipt.	1.The system calculates total bill on the chosen products. 2. Calculates the final total, generates a detailed bill, and presents it for confirmation. 3.Upon the user's request, the system processes the transaction data and generates a detailed receipt.

Use Case Number 03	Transactions record	
ID:	1.3	
Actors Involved:	Shopkeeper	
Brief Description:	The system records details of each transaction, including date, time, items purchased, quantities, individual prices, applicable discounts, and the total amount. This information serves for accounting, auditing purposes, and customer reference.	
Pre-Conditions:	**Product Database Update**: The system's product database must be up-to-date. Product information, including prices, is accurate and current. **Completed Transaction**: All necessary steps of the transaction, including product selection, billing, and payment, are finalized.	
Post-Conditions:	**Transaction Record Creation:** A detailed transaction record is created and stored in the system. **Integration with Accounting System:** Transaction details are integrated into the accounting system for accurate financial records.	
Normal Flow of Events:	**Actor Action**	**System Response**
	1.The user (shopkeeper or customer) completes the necessary steps of a transaction, including product selection, review, and confirmation. 2.The user reviews the generated receipt or transaction record.	1.After a successful transaction, the system creates a detailed record, capturing all relevant information. 2. For online transactions, the system sends communication containing the transaction details to the customer. Simultaneously, it integrates the transaction details into the accounting system.

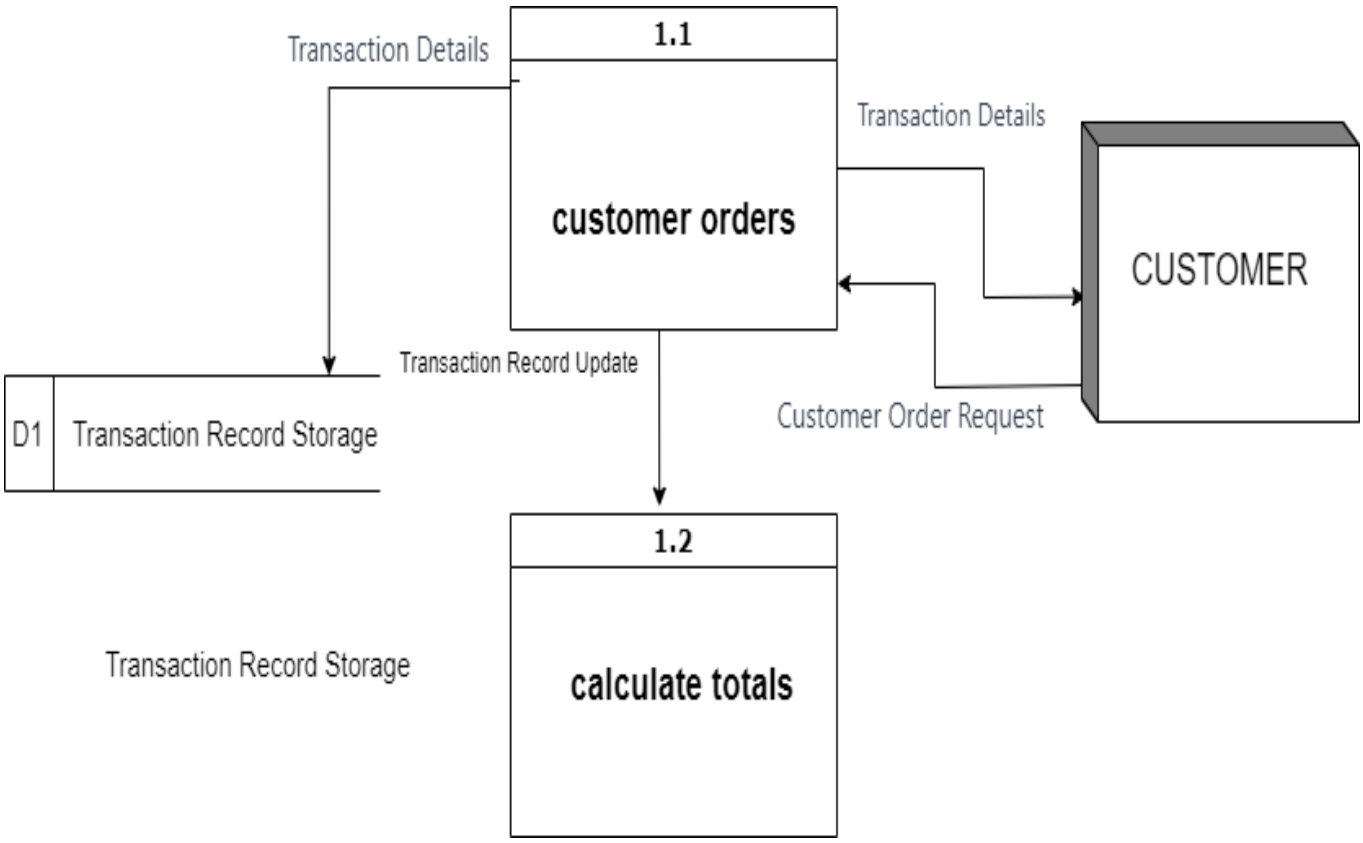
CONTEXT DIAGRAM:



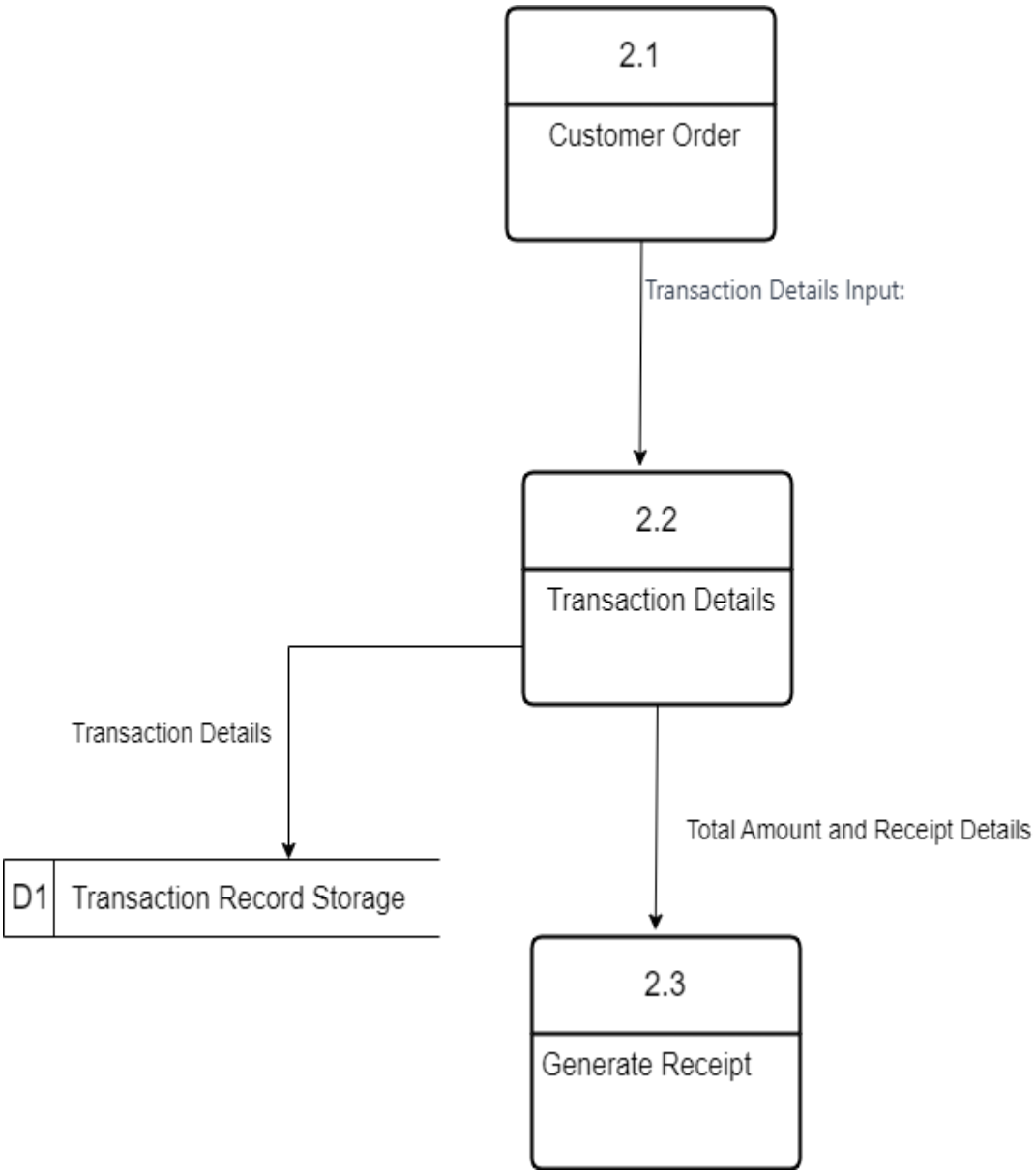
DFD DIAGRAM (0):



DFD DIAGRAM (1):



DFD DIAGRAM (2):



Suggested Model:

We suggested the Agile Process Model allows for flexible and iterative development.
Given that the digital canteen billing system might require frequent adjustments based on user feedback or changing business needs, Agile provides the flexibility to adapt to these changes.

References:

https://www.scribd.com/document/410537515/SRS-FOR-CANTEEN-MANAGEMENT
https://www.scribd.com/presentation/527036703/Canteen-Management-System2

THANK YOU!